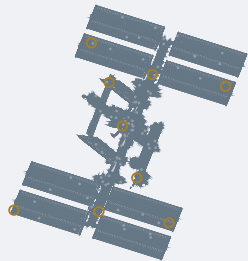


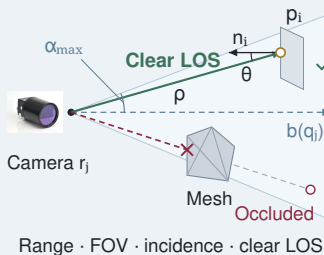
Viability-preserving rollout ADP for orbital inspection

a Mission inputs



○ Required targets K

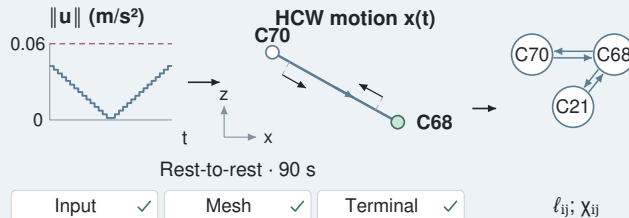
b Camera and LOS



Valid targets $\rightarrow G_i$

$$s = (j, m, b, h)$$

c HCW transfer $\rightarrow$ SOOA graph



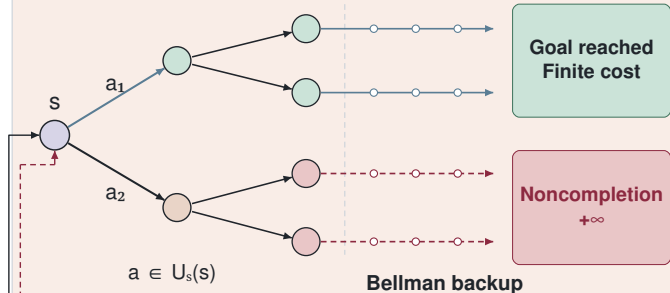
d Rollout ADP

Finite Bellman prefix

Lookahead depth d

Task-aware policy tails

Deterministic greedy μ



$$\widehat{Q}_d(s, a) = l_{ja} + \widehat{V}_{d-1}(f(s, a))$$

State update and replan

e Replanning

First action a^*

Minimum finite Q

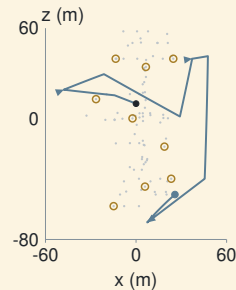
Append arc + update

$$s' = f(s, a^*)$$

Required goal reached?

Yes No

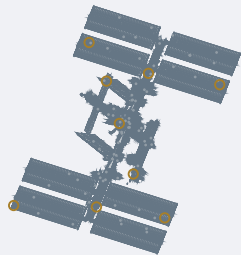
Planned route



Required targets: 9/9

Orbital inspection: sensing, audited transfers and rollout ADP

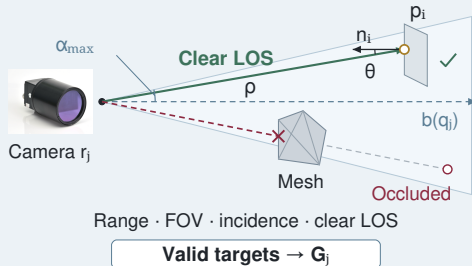
a Mission



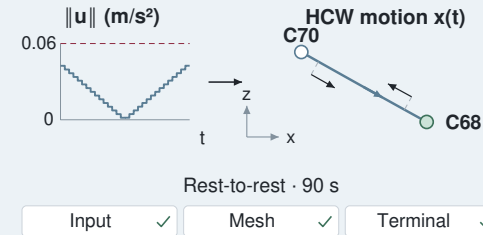
○ Required targets K

Initial state; budget H

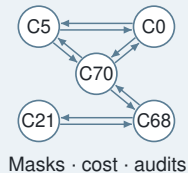
b Camera and LOS



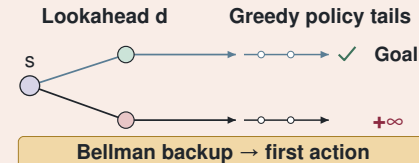
c Controlled HCW transfer



d SOOA graph



e Rollout ADP



f Inspection plan



Rollout ADP with task-aware base-policy evaluation

a Decision state

$$s = (j, m, b, h)$$

j current viewpoint
m covered targets
b selected views
h remaining budget

Audited + unvisited

$$\mathcal{U}_s(s): b_a = 0, \chi_{ja} = 1$$

Safe connecting
actions retained

Fixed required set K

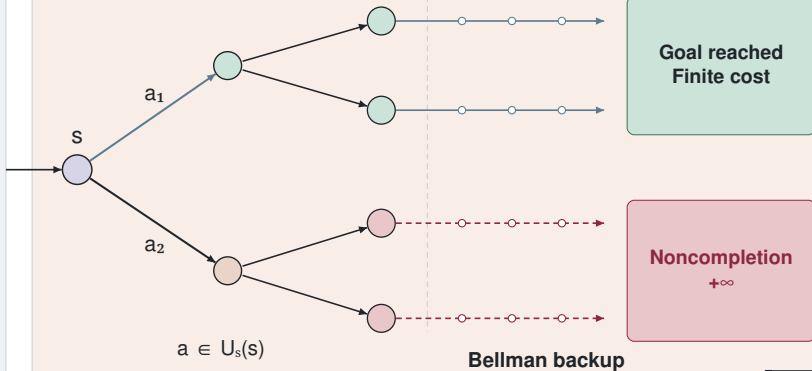
b Rollout ADP

Finite Bellman prefix

Lookahead depth d

Task-aware policy tails

Deterministic greedy μ



All $Q = +\infty \rightarrow$ no certified completion

c Policy update

Finite values only

$$a^* = \arg \min_a \hat{Q}_d(s, a)$$

Append first arc and update

$$s' = f(s, a^*)$$

Required goal reached?

Yes No

Complete plan

State update and replan